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Increased anterior pelvic tilt angle elevates the risk of hamstring injuries in soccer player

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ABSTRACT

Objective: This study aims to determine the impact of increased anterior pelvic tilt angle on hamstring muscle injuries in football players. An independent t-test was performed to compare right and left anterior pelvic tilt angles.

Design: The study was conducted by monitoring 76 different players from a professional club in the Premier Division of Professional Football League over a period of five years. During this time, the hamstring injuries sustained by the players and their severity were recorded using MRI imaging techniques.

Result: The comparison of the right and left anterior pelvic tilt angles among players was analysed using an independent t-test for independent groups. The Receiver Operating Characteristic (ROC) analysis was used to determine the threshold value that categorizes the right and left angles. According to the results, a significant impact of the variation in right-left anterior pelvic tilt angle on hamstring muscle injuries was observed ($p < 0.05$). However, no significant effects were found for the Q angle, tibial torsion, or knee posture ($p > 0.05$).

Conclusion: In conclusion, it can be stated that an increase in the anterior pelvic tilt angle is associated with a higher frequency of hamstring injuries.

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KEYWORDS
Anterior pelvic tilt angle; hamstring injuries; soccer

Introduction

The hamstring muscles consist of three main muscles located on the posterior thigh: the *Semimembranosus-Semitendinosus* (ST-SM) located medially and the laterally situated long and short heads of the *Biceps Femoris* (BF). These muscle groups are clinically significant due to their high susceptibility to injuries, especially in athletes (Biz et al., 2021).

Hamstring injuries (HI) are commonly observed in sports, with a general incidence rate of 1.2–4 per 1,000 training sessions (Roe et al., 2016). In athletics, they account for 17–21% of total injuries(Edouard et al., 2016), while approximately 22% of soccer players are reported to sustain an HI each season (Ekstrand et al., 2016). HIs typically result in an average of 24 days lost from training, imposing significant costs on professional athletes and teams (Ahmad et al., 2014; Hickey et al., 2014; Kamandulis et al., 2023). Thus, preventing HIs in sports is of great importance (Danielsson et al., 2020). These injuries

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frequently occur in sports that involve stretching, jumping, sudden turns, and high-speed running. As with all muscle injuries in football, HI injury rates are higher during matches than during training (Jokela et al., 2023).

Most hamstring injuries occur in the BF located in the lateral region, often during sprinting activities (Arnason et al., 1996; Brooks, Fuller, et al., 2005; Jokela et al., 2023; Woods et al., 2004). The literature on HIs focuses on three main mechanisms. First, the force deficiency mechanism suggests that low muscle strength or fatigue reducing eccentric hamstring force output increases the risk of injury, and decreased endurance post-injury raises the risk of recurrence (Jones et al., 2015; Schuermans et al., 2016; Woods et al., 2004). Second, strain-related HIs occur because hamstring muscles span two joints, and movements involving both joints can lead to excessive muscle tension, potentially causing muscle damage during simultaneous knee extension and hip flexion (Danielsson et al., 2020; Jokela et al., 2023). Third, injuries related to running biomechanics, also referred to as mixed type, occur under the stresses of force and tension (Danielsson et al., 2020; Jokela et al., 2023). However, further research is needed to validate these mechanisms and their specific contributions to hamstring injury risk in sports involving high-intensity movements (Danielsson et al., 2020).

Improper posture can lead to musculoskeletal pathologies, with pelvic position playing a crucial role in many postural defects (Sahrmann, 2002). Therefore, assessing standing pelvic posture in clinical practice is essential for identifying potential issues with the spine and peripheral joints (Youdas et al., 1996). Among the most common anatomical variations in soccer players is genu varum (Isın & Melekoğlu, 2020; Thaller et al., 2018; Witvrouw et al., 2009), which can be exacerbated by intense football training, irregular stresses, torsional forces, and traumas on the knee joint (Isın & Melekoğlu, 2020). Although increased Q angle is associated with anterior knee pain and patellar instability (Rougereau et al., 2023; Thaller et al., 2018), the literature lacks information on its relation to muscle injuries and specifically HIs.

Anterior pelvic tilt (APT) is measured as the angle between the line connecting the anterior superior iliac spine (ASIS) and the posterior superior iliac spine (PSIS), with a normal reference value in the healthy population of approximately 8 degrees (Brekke et al., 2020; Herrington, 2011; Nguyen & Shultz, 2007). Although no fixed cut-off point for excessive APT is established in the literatüre (Brekke et al., 2020; López-Miñarro et al., 2012; Ludwig et al., 2016), a higher PSIS indicates a greater degree of APT (Freburger & Riddle, 1999; Herrington, 2011).

Asymmetries in pelvic structures are known to lead to a series of postural compensations, predisposing individuals to numerous neuromusculoskeletal dysfunctions (Juhl et al., 2014). Retroversion increases pelvic tilt (PT) and accompanying lordosis, whereas anteversion reduces PT and consequently, lordosis (Le Huec et al., 2011). An increase in APT along with increased lordosis intensifies tension in the hamstring muscles (Hanson et al., 2023), and likewise, hamstring tension can alter the APT (Muyor et al., 2011). Studies have found a correlation between hamstring tension and APT, but more research is needed to link APT with hamstring injuries directly. Moreover, no valid cut-off point for excessive anterior pelvic tilt has been established due to variations in population-based normative data, the lack of consistent measurement standards, and limited literature (Brekke et al., 2020; Roussouly et al., 2005).

Despite numerous studies on the aetiology and mechanisms of HIs and identifying potential risk factors, there has been no improvement in the frequency and burden of injuries (Crawford et al., 2023; Iatropoulos & Whe Wheeler, 2024). Therefore, identifying risk factors for hamstring injuries and developing prevention programmes require a different perspective under biomechanical differences. The hypothesis of the study is that changes in the anterior pelvic tilt angle will increase the occurrence of hamstring injuries. This study aims to elucidate the relationship between APT and hamstring injuries and to determine the angular magnitude at which HIs are triggered.

Methods

Participants

The study was conducted over a five-year period with players from a club in the Premier Division of Professional Football League. Participants included first-team players who had been playing professional football for at least five years (Table 1). During this period, hamstring injuries sustained by 76 different football players were recorded (flowchart).

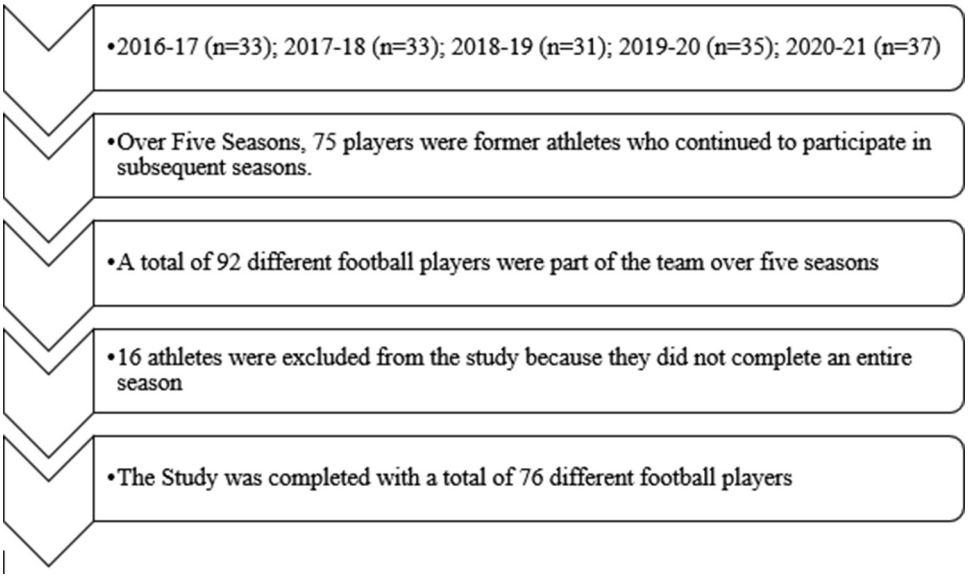


Table 1. Demographic information of participants.

	Hamstring Injury		
	Injuried ($n = 33$)	Non Injuried ($n = 43$)	t value
Age (years)	24.67 ± 4.13	26.21 ± 3.93	1.026
Body Weight (kg)	78.47 ± 3.82	78.81 ± 4.30	0.278
Height (m)	1.81 ± 0.063	1.83 ± 0.07	1.207
BMI (kg/m^2)	24.13 ± 2.15	23.88 ± 1.86	−0.957

BMI: Body Mass Index, kg: kilogram, m: metre.

Players who were sidelined from training for at least three days due to a hamstring injury, as identified by the club's sports physician, were included in the study. The severity of injuries was documented using MRI imaging methods. Players who suffered a hamstring injury during rehabilitation from another injury and players under 18 who experienced a hamstring injury during first-team training sessions were excluded from the study. Ethical approval for the study was obtained from the X University Faculty of Sports Sciences on 4 April 2024, under the approval number 2024/41. Each participant was informed about the procedures involved in the study and provided informed consent. Additionally, written permission was obtained from the football club to use the data collected for scientific research.

The eta squared (η^2) method was used to determine the effect size via power analysis. Eta squared (η^2) values were obtained as 0.37 and 0.36, which fall within the range of a large effect size according to Cohen's general effect size standards.

Data collection

APT measurement

All measurements were conducted by a single experienced physiotherapist. The athletes stood with equal weight on both feet. APT was measured using a caliper pelvic tilt metre. Each subject stood on a platform with a width of 30 cm, and to control for postural sway, they were instructed to focus on a fixed point in front of them. The anterior superior iliac spine (SIAS) and posterior superior iliac spine (PSIS) points were identified and marked. The caliper tips were placed directly on these marked points, and the angle was measured with an inclinometer and recorded. Three measurements were taken for both sides to obtain an average (Gajdosik et al.,1985; Hagins et al., 1998; Krawiec et al., 2003). Studies have shown that caliper-based pelvic inclinometers are a valid and reliable method. Additionally, they have demonstrated acceptable validity when compared to radiographs for assessing the amount of anterior pelvic tilt and total pelvic tilt (Hayes Ann et al., 2016; Suits, 2021).

Knee posture evaluation

Football players were assessed in a standing upright position to check for the presence of genu varum and genu valgum. Players were instructed to touch their medial malleoli together, and the distance between their knees was measured. A gap of 0–3 cm was considered normal. A gap greater than 3 cm indicated the presence of genu varum. In cases where the players could not bring their medial malleoli together due to contact at the medial part of the knee, genu valgum was considered present (Soheilipour et al., 2020).

Q angle measurement

Players were positioned supine with their knees extended. The spina iliaca anterior superior (SIAS) and tibial tuberosity were identified. It was ensured that the patella was positioned within the trochlear groove. A goniometer was used with its distal arm over the tibial tuberosity and the proximal arm extended towards the SIAS. The measurement was completed as the other arm of the goniometer was extended towards the SIAS (Merchant et al., 2020)

Tibial torsion measurement

Players lay prone with their knees flexed at 90 degrees. The medial and lateral malleoli's midpoint was marked, and a line was drawn to connect these points. Another line perpendicular to this line was drawn, and the angle between this line and the midline of the thigh was measured with a goniometer (Singh et al., 2022).

Statistical analysis

The normality of the angle distributions for the groups was tested using Skewness-Kurtosis and Shapiro-Wilk normality tests. Independent t-tests were used for comparing the normally distributed right and left APT in independent groups. ROC analysis was employed to determine the threshold value that categorizes the right and left angles into binary classifications (injury present/no injury). The cut-off values determined based on the highest Youden index and the AUC (Area Under the ROC Curve) values are presented in Table 9. The level of statistical significance was set at $p < 0.05$. Statistical values were calculated using JAMOVI version 2.4.8.

Results

It was found that there were no significant differences in age, weight, height, and body mass index (BMI) between the groups of football players who experienced hamstring injuries and those who did not ($p > 0.05$). This suggests that these demographic factors did not significantly influence the occurrence of hamstring injuries in this sample (Table 1).

33 football players have experienced a total of 66 hamstring injuries. The distribution of hamstring injuries demonstrates that the muscle belly was most commonly affected, with 54.5% of injuries occurring in this region. Proximal tendon injuries accounted for 9.1% of cases involving the muscle belly, and only one case was recorded as an isolated proximal tendon injury. Distal hamstring tendon involvement was observed in 21.2% of the cases, while combined injuries involving both the muscle belly and distal tendon occurred in 6.1% of the cases. This breakdown provides insights into the common sites and combinations of hamstring injuries among athletes (Table 2).

The majority of injuries (72.7%) were found to occur during matches, while approximately a quarter (27.3%) originated from training sessions. This distribution highlights the increased risk associated with competitive play compared to training environments (Table 3).

Table 2. Anatomical tissue location of hamstring injuries.

	Hamstring injury	
	Number of injuries	Percentage
Muscle Belly	36	% 54.5
Distal Hamstring Tendon	14	% 21.2
Proximal Hamstring and Muscle Belly	10	% 15.2
Muscle Belly and Distal Hamstring Tendon	4	% 6.1
Proximal Hamstring Tendon	2	% 3.0
Total	66	% 100

Table 3. Time of occurrence of hamstring injuries.

	Hamstring injury	
	Number of injury	percentage
During Training	18	%27.3
During Match	48	%72.7
Total	66	%100

Table 4. Distribution of hamstring injuries by muscle.

	Hamstring injury	
	Number of injury	Percentage
Medial Hamstring (Semitendinosus- Semimebranosus muscle)	16	% 24.2
Lateral Hamstring (Biceps Femoris Muscle)	50	% 75.8
Total	66	% 100

The distribution of hamstring injuries revealed that the BF was the most commonly affected muscle, with 75.8% of injuries occurring in this region. The medial hamstring group, which includes the ST and SM, accounted for 24.2% of the injuries. This data underscores the BF as a particularly vulnerable muscle to injury in the context of hamstring-related sports injuries (Table 4).

Right and left anterior pelvic tilt, Q angle, and tibial torsion were tested for normality using Skewness-Kurtosis and Shapiro-Wilk tests. To determine whether these factors significantly influenced hamstring muscle injuries BF and ST-SM, multiple linear regression analysis was performed. The results of the regression analysis indicated that the model was significant ($p = .00$). The explanatory power (R^2 value) of right APT was calculated as 0.37 ($R = 0.608$, $R^2 = 0.37$, $p < 0.05$). For left APT, the R^2 value was 0.361 ($R = 0.601$, $R^2 = 0.361$, $p < 0.05$). Since $p < 0.05$ for both right and left APT, it was determined that these factors have a significant effect on hamstring muscle injuries (Table 5).

However, no significant effects were found for the Q angle, tibial torsion, or knee posture. It was noted that as the Q angle decreases, there is a slight influence on hamstring injuries (right Q angle .155, left Q angle .117), and a very weak effect of knee posture (.442). No significant impact was observed for right tibial torsion (.759) and left tibial torsion (.854).

Table 5. Multiple regression analysis of factors potentially causing hamstring injuries.

	B	Std. Error	ß	t	*P*
Right APT	393	.164	.608	2.388	0.00
Left APT	418	.164	.601	2.552	0.00
Right TT	1.391	.152	.036	9.167	.759
Left TT	1.407	.156	.022	9.045	.854
Right Q Angle	1.193	.177	.165	6.731	.155
Left Q Angle	1.166	.179	.181	6.522	.117
Knee Posture (genu varum-valgum)	1.481	.083	−.089	7.758	.442
Age	1.800	.366	−.117	4.920	.315
Weight	1.740	1119	−.032	1.555	.785
Height	3.294	1.546	−.139	2.131	.233

APT: Anterior pelvic Tilt, TT: Tibial Torsion

Additionally, while the ages, weights, and heights of the athletes were not found to significantly affect injuries, it was determined that age (.315) and height (.233) have a somewhat more pronounced effect compared to weight (.785).

The explanatory power (R^2 value) for knee posture on biceps femoris muscle injuries was calculated to be 0.020 ($R = 0.140$, $R^2 = 0.020$, $p = 0.226$), indicating a very weak effect, although not statistically significant. It was determined that 56% of hamstring injuries occurred in individuals with normal knee posture, 36% in those with genu varum, and 8% in those with genu valgum (Table 6).

While the effect of knee posture on hamstring injuries was quantified with a p-value of 0.442, its effect on the biceps femoris muscle specifically was found to be $p = 0.226$, suggesting a somewhat stronger impact on the biceps femoris than on hamstring injuries in general, although still not reaching statistical significance.

The explanatory power (R^2 value) of the right Q angle on biceps femoris muscle injuries was calculated to be 0.012 ($R = 0.108$, $R^2 = 0.012$, $p = 0.354$), indicating a very weak effect. For the left Q angle, the explanatory power on the biceps femoris was slightly higher at an R^2 value of 0.018 ($R = 0.135$, $R^2 = 0.018$, $p = 0.244$). Although these effects are not statistically significant, they are present, albeit weak.

When comparing the influence of the Q angle on hamstring injuries and specifically on the biceps femoris muscle, it was noted that the effects on hamstring injuries were stronger. The p-values for the right and left Q angles' impact on hamstring injuries were p = 0.155 and p = 0.117, respectively, whereas their effects on the biceps femoris were p = 0.354 and p = 0.244, respectively. This suggests that the Q angle has a greater impact on hamstring injuries in general than on injuries to the biceps femoris specifically (Table 7).

According to Table 8, it was seen that the right and left APT values of the players with HI were greater than the APT values of the players without injury. ($z = 8.907$, $z = 9.267$, $p < 0.05$, respectively). It was determined that the risk of HI increased as the APT increased (Table 8).

Table 6. Relationship between knee posture (Genu Varum, Genu Valgum) and isolated biceps femoris injuries.

	B	Std. Hata	ß	t	R	R^2	F	*P*
Knee posture	.926	.154	–.140	.6104	.140	.020	.1489	.226

Table 7. Q angle relationship between isolated biceps femoris injuries.

	B	Std. Error	ß	t	R	R^2	F	*P*
Right Q Angle	496	.332	.108	1.492	.108	.012	.869	.354
Left Q Angle	416	.335	.135	1.242	.135	.018	.1380	.244

Table 8. Anterior pelvic tilt relationship between hamstring injury.

	Hamstring injury			
	Injuried (n = 33)	Non injuried (n = 43)	z value	P value
Right APT(°)	14.12±3.00	9.72±2.235	*8.907*	0.001
Left APT(°)	14.20±2.94	9.69±2.32	*9.267*	0.001

APT: Anterior Pelvic Tilt

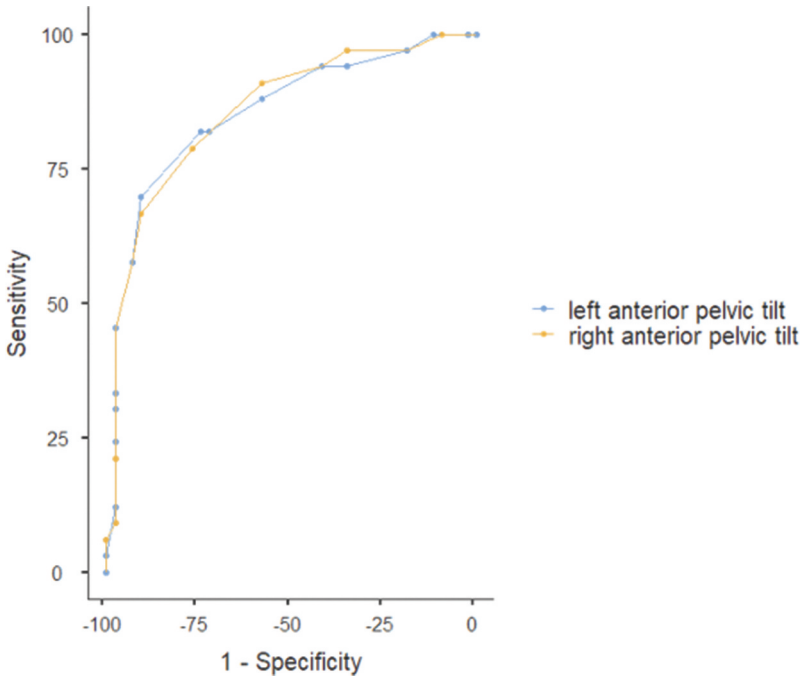
Table 9. APT and cut-off values according to injury status.

	AUC	Z value	P	Standart error	95% CI	Sensitivity	Specificity	Cut-off	Youden index
Right APT	0.87	8.907	<0.001	0.042	0.779-0.940	69.70	93.02	≥13	0.627
Left APT	0.88	9.267	<0.001	0.041	0.798-0.953	72.73	93.02	≥13	0.657

APT:Anterior Pelvic Tilt, AUC: Area Under the Curve, ROC: Receiver Operating Characteristic **According to ROC analysis, the cut-off and AUC values were obtained. The cut-off points were determined based on the highest Youden index value.*

ROC (Receiver Operating Characteristic) analysis is a valuable analytical method in situations where the diagnostic process is time-consuming, costly, and requires specialized methods, equipment, and skilled personnel (Kılıç, 2013). It facilitates the clinical decision-making process by identifying appropriate cut-off points for easily obtainable markers quickly and at low cost. Scores obtained through linear, quadratic, and logistic separation analyses were placed into both parametric and non-parametric ROC curve methods separately. This process allowed for the calculation of the area under the ROC curve (AUC), the optimum cut-off point, sensitivity, specificity, and Youden index values. The selection of the optimal cut-off point, considering sensitivity and specificity, was determined by the maximum Youden index value (Bilgin et al., 2020).

For the right anterior pelvic tilt (APT), a cut-off value of 13° was identified based on an AUC value of 0.87 and a Youden index result of 0.627. Similarly, for the left APT, a cut-off value of 13° was determined with an AUC value of 0.88 and a Youden index of 0.657 (Table 9). It was found that APT of 13° and above are associated with an increased risk of injury. This analysis underscores the critical role of determining specific thresholds that can effectively predict risk levels in clinical assessments.



Graphic 1. Roc curve.

In ROC (Receiver Operating Characteristic) analysis, the closer the area under the curve (AUC) is to 1, the higher the diagnostic value is considered. The most optimal positive cut-off value is the point on the curve closest to the top-left corner of the graph. For right anterior pelvic tilt (APT), a sensitivity of 69.70 and a specificity of 93.02 were identified as the point nearest to the top-left corner. For left APT, a sensitivity of 72.73 and a specificity of 93.02 were determined as the closest point to the top-left corner (Graphic 1). In an ROC curve, an area value of 1 corresponds to 100% diagnostic power.

For both right and left APT, the findings from the cut-off values, as the curve approaches 100 sensitivity in the top-left corner, indicate a strong diagnostic value. This positioning underscores the robustness of the identified thresholds in effectively predicting the risk of hamstring injuries based on APT measurements.

Discussion

This study aimed to investigate whether anterior pelvic tilt (APT) has an effect on hamstring injuries. In our research, the effect of lower extremity posture on hamstring injuries was examined, and it was found that a lower APT angle increases hamstring injuries. Additionally, in our study, which examined football players over 5 seasons, the general profile of hamstring injuries was revealed.

It was significant that the age, height, weight, and BMI values of athletes with and without hamstring injuries were similar, enhancing the validity and reliability of our study. One of the strengths of our research is the pre-injury documentation of football players' APT. The average APT angles measured were 14.54 ± 2.87° for both right and left sides. In comparison, a study on sedentary males found average APT of 6.74° on the left and 6.23° on the right, with some individuals showing a posterior pelvic tilt (Herrington, 2011). Another study reported a pelvic tilt of 10.7 ± 9.0° (Imai et al., 2019). Athletes, particularly football players, tend to have higher anterior pelvic tilt angles than non-athletes, as one study measured the APT at 18.7 ± 2.4° in football players (Alızadeh & Mattes, 2019). However, that study noted that during running, the APT varied between 8 to 25 degrees.

In line with existing research indicating that the majority of hamstring injuries (HIs) occur during competitive matches (Ekstrand et al., 2022; Moreno-Perez et al., 2023), our study similarly found that 72.7% of HIs happened during matches, while 27.3% occurred during training sessions. This distribution emphasizes the heightened risk of injury during competitive play, possibly due to the higher intensity and aggressive nature of match conditions compared to training.

The findings indicate that the BF muscle is particularly susceptible to injuries, which aligns with previous studies reporting that 70% of HIs occur in the BF region (Grange et al., 2023). This concurrence supports the reliability of our results and contributes further to the understanding of injury distribution within the hamstring group, highlighting the biceps femoris as particularly susceptible to injuries in athletic activities.

The study by Gudelis et al. published in 2024, which tracked athletes from 17 different sports at Barcelona Sports Club over a decade, reported injury rates in hamstring muscles as follows: 45.8% in the BF, 32.1% in the SM, and 22.1% in the ST. The lower incidence rate for BF compared to some literature might be attributed to the diversity of the study's participants, which included both male and female athletes across various sports disciplines. Despite this variation, injuries to the BF muscle were still the most frequent

(Gudelis et al., 2024). This broader scope and inclusion might explain the variance in injury distribution, offering insights into how different sports and gender considerations can impact the prevalence and nature of hamstring injuries.

In their 2024 histomorphological study, Kim and colleagues reported that the BF has a collagen content similar to that of the ST, but it also contains a significant amount of multidirectional elastic fibres. The ratio of elastic fibres to collagen in these muscles can affect the muscle's contraction angle, potentially leading to injuries due to differences in eccentric loading forces between the BF and ST (Kim et al., 2024). Despite both being proximally originated and having integrated structures, the reasons for the BF's higher susceptibility to injury during muscle contraction activities remain a topic of debate.

The BF is characterized by strong contractility across most biomechanical parameters, yet it is particularly sensitive to instantaneous eccentric forces, which challenge the muscle structure's resilience (Ono et al., 2015). This sensitivity to sustained and higher intensity eccentric stress compared to the ST places the BF at a greater risk for potential injuries (Ono et al., 2015). Conversely, the SM contains a high proportion of type 1 collagen fibres, giving it a robust capacity to withstand repetitive external strain forces due to its high collagen content (Kim et al., 2024). This structural composition makes the SM less prone to injuries compared to the BF under similar conditions. This distinction in tissue composition and biomechanical properties across hamstring muscles provides insight into their differing vulnerability to injuries and the potential need for targeted prevention strategies specific to each muscle's characteristics.

The BF is the only hamstring muscle anatomically attached to the ischial tuberosity via the sacrotuberous ligament (Pérez-Bellmunt et al., 2015; Woodley & Mercer, 2005). This unique anatomical connection makes the BF particularly susceptible to increased tension during hip flexion, which can result from increased pelvic tilt (PT) caused by tension in the iliopsoas muscle. Consequently, BF injuries are observed more frequently (Chumanov et al., 2007; Franz et al., 2009; Visser et al., 1990).

Furthermore, while reductions in the Q angle and genu varum have not been found to directly affect hamstring injuries overall, our findings suggest a potential low-level association between these factors and isolated BF injuries. This might be due to the anatomical tension created in the BF due to the genu varum condition. This condition places significant stress on the iliopsoas, which in turn can disproportionately affect the BF located laterally (Lucha-López et al., 2018). Despite previous studies on the effects of genu varum on various muscles (Letafatkar et al., 2020; Marks et al., 1994), there is a notable gap in the literature regarding its specific relationship with hamstring muscles among football players, who commonly exhibit this knee alignment disorder.

This gap highlights the need for further research to explore how genu varum and other biomechanical factors specifically influence the BF and other hamstring muscles, considering their unique anatomical and functional roles in athletic movements. Such studies could provide deeper insights into tailored prevention and treatment strategies for athletes prone to specific alignment disorders.

Our study aimed to investigate the impact of the APT on the incidence of hamstring injuries. We found that an increase in the APT was associated with an increased risk of hamstring injuries. This is consistent with the notion that changes in the APT can alter the femoral head's placement in the acetabulum, potentially affecting the muscles attached to the femur (Hodel et al., 2023; Hong et al., 2023; Okuzu et al., 2019).

HI, though typically occurring unilaterally during sprinting, involve both extremities interacting anatomically and biomechanically with the pelvis, making the pelvis a key element in the comprehensive scheme of sprint-related HIs (Mendiguchia et al., 2021). During sprinting, an increased APT can further enhance the lordosis angle (Franz et al., 2009). This explains the link between increased APT and lordosis and the subsequent rise in hamstring injuries in sports involving running (such as football and athletics) compared to other sports (like cycling and swimming) (Bloomfield & Ellito, 1998; Edouard et al., 2016; Kritz & Cronin, 2008; Watson, 1993; Wodecki et al., 2002). Trigger events like linear acceleration, high-intensity running, braking, or stopping, and kicking are associated with hamstring tension (Gregson et al., 2020; Moreno-Perez et al., 2022).

A larger APT increases the demand on hamstring strain due to a larger moment arm. It also compensates for the lack of increase in hip extension required to reach high speeds, which is a fundamental requirement for sprinting. While this scenario provides an advantage in running speed for athletes, it places the hamstring muscles under constant stress, posing a disadvantage (Chaudhari et al., 2014; Daly et al., 2016; Schuermans et al., 2017; Nagahara et al., 2018).

The postural changes observed between individuals who have and have not suffered from HI provide insights into the nature of HIs (Nurse et al., 2023). Interventions to adjust the APT could potentially prevent hamstring injuries (Mendiguchia et al., 2020), supported by our findings that an increase in the APT is associated with a higher risk of hamstring injuries, aligning with the literature.

In their 2024 study, Mendiguchia and colleagues designed an apparatus to adjust the femur and pelvis bones on seven cadavers, thereby increasing APT. This adjustment led to a significant increase in muscle tension across all muscles in the hamstring group. They found that changes in the APT angle could potentially cause hamstring injuries, highlighting the biomechanical impact of pelvic orientation on muscle strain. This study provides valuable insights into how pelvic alignment may contribute to the risk of hamstring injuries (Mendiguchia et al., 2020).[1]

Preventive measures for HIs have been supported by a meta-analysis by (Rudisill et al., 2023), which highlighted that stretching protocols to increase hamstring flexibility are crucial for preventing these injuries (Rudisill et al., 2023). Another meta-analysis on HI treatment emphasized the significant effectiveness of stretching exercises in treatment (Poursalehian et al., 2023). While research has explored muscle strength, muscle imbalance, and flexibility as modifiable factors influencing hamstring injuries, results have been mixed (Kamandulis et al., 2023; Mendiguchia et al., 2024; Poursalehian et al., 2023; Rudisill et al., 2023). However, the consensus suggests that improvements in flexibility and muscle strength can enhance the efficacy of injury treatment.

Our study adds that the biomechanical effect directly impacting muscle strength and flexibility contributes to increased injury rates, indicating that flexibility and muscle strength should be considered inconjunction with the APT. This finding is significant because it underscores the need to evaluate these factors together when assessing the risk of hamstring injuries.

Furthermore, considering the APT alongside these modifiable risk factors may enhance the significance of study outcomes. The literature's conflicting results on these factors can gain clarity and provide more definitive guidance for prevention and treatment strategies when analysed with the biomechanical context provided by the APT. This holistic

approach could potentially lead to better targeted interventions aimed at reducing the prevalence of hamstring injuries in athletes.

Our study also contributes to the literature by being among the first to establish a cut-off value for APT in relation to hamstring injuries in football, suggesting that this measure could be crucial for assessing the risk of hamstring injuries and developing injury prevention programmes. The conflicting results in the literature regarding modifiable risk factors, when considered alongside APT, might show increased levels of significance. This suggests that APT should be evaluated in conjunction with flexibility and muscle strength as important variables in managing and preventing hamstring injuries.

Limitations

Our study has several limitations. A primary constraint is its retrospective nature, which complicates the establishment of causality. The study included only male football players, which, while important for standardization, limits the diversity of cases and the assessment of potential gender-specific effects. Additionally, our analysis based on knee and pelvic posture did not incorporate parameters such as muscle strength, representing another limitation. Furthermore, the study was not applied across different sports disciplines or a broader population, limiting the generalizability of the findings.

Conclusion

Our study, which followed professional football players over five years, demonstrated that an increase in the anterior pelvic tilt angle (APT) is associated with a higher frequency of hamstring injuries. Given the high incidence of such injuries in football, a sport that frequently involves running and changing direction, the APT should be considered a significant risk factor. We believe our findings contribute to the literature by highlighting the importance of the APT in lower extremity alignment and suggesting that an APT exceeding certain values increases the likelihood of hamstring muscle injuries in football players. We strongly emphasize that including players with an APT above 13° for both right and left pelvis in hamstring injury prevention programmes could reduce the incidence of such injuries and that the APT should be considered a critical criterion in pre-season evaluations.

Note

1. Mendiguchia, J., Conceição, F., Edouard, P., Fonseca, M., Pereira, R., Lopes, H., ... & Jiménez-Reyes, P. (2020). Sprint versus isolated eccentric training: Comparative effects on hamstring architecture and performance in soccer players. Plos one, 15(2), e0228283.

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